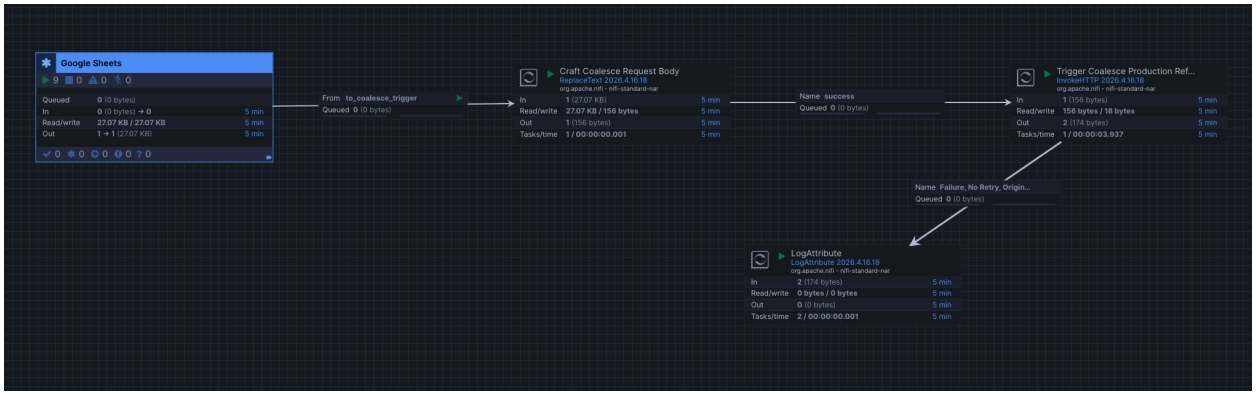
Configuring Coalesce Transform Trigger in OpenFlow Runtime

This guide outlines the steps to replicate the Coalesce Job Trigger logic within an **Openflow (Apache NiFi)** runtime. This pattern ensures that a Coalesce Data Transformation Graph is triggered automatically once a source ingest (in this case, MySQL) successfully concludes.



Architecture Overview

The trigger chain consists of three primary components appended to the end of the ingest module:

1. **ReplaceText (`Craft Coalesce Request Body`):** Formats the API instructions.
2. **InvokeHTTP (`Trigger Coalesce Production Refresh`):** Executes the API call to Coalesce.
3. **LogAttribute:** Captures the API response for auditing and troubleshooting.

Step 1: Module Integration (The "Success" Signal)

To trigger the job only after the MySQL data has successfully landed in Snowflake:

1. **Inside the MySQL Module:** Navigate to the final processor (typically a `PutSnowflake` or `PutSQL` variant).
2. **Add an Output Port:** Drag an **Output Port** onto the canvas inside the group and name it `to_coalesce_trigger`.
3. **Connect:** Link the **`success`** relationship of the final insert processor to this Output Port.
4. **On the Main Canvas:** Connect the **MySQL Process Group** to the first component of the trigger chain (`ReplaceText`).

Step 2: Configure "Craft Coalesce Request Body" (ReplaceText)

This processor wipes the ingest metadata and replaces it with the Coalesce API payload.

- **Replacement Strategy:** Always Replace
- **Evaluation Mode:** Entire Text
- **Replacement Value:**

JSON

None

```
{
  "runDetails": {
    "environmentID": "25",
    "parallelism": 16,
    "jobID": "47"
  },
   "userCredentials": {
    "snowflakeAuthType": "KeyPair"
       }
}
```

> **Note:** The environmentID is the numerical ID found in the Coalesce URL when viewing the target environment.

Step 3: Configure "Trigger Coalesce Production Refresh" (InvokeHTTP)

This acts as the HTTP operator to communicate with the Coalesce Scheduler.

Properties

Property	Value

HTTP Method	POST
HTTP URL	https://app.coalescesoftware.io/scheduler/startRun
Content-Type (Dynamic Property)	application/json
Authorization (Dynamic Property)	Bearer <YOUR_COALESCE_API_TOKEN>

*To add the Authorization header, click the **+** icon in the properties tab, name it Authorization, and ensure the value starts with Bearer followed by the token.*

Relationships

- **Auto-terminate:** Original, No Retry, Retry.
- **Connected:** Link Response and Failure to the **LogAttribute** processor.

Step 4: Configure LogAttribute

This processor is essential for confirming the API accepted the request.

- **Log Level:** info
- **Log Payload:** true (Optional, helpful for debugging the JSON response from Coalesce).
- **Relationships:** Auto-terminate success.

Step 5: Scheduling & Execution

To ensure the process runs predictably:

1. **Ingest Frequency:** Set the **Run Schedule** on the *first* processor of the MySQL module (e.g., ListDatabaseTables or QueryDatabaseTable) to the desired interval (e.g., 5 mins).
2. **Trigger Chain Frequency:** Set the **Run Schedule** for both ReplaceText and InvokeHTTP to **0 sec**. This ensures they execute immediately upon receiving the signal from the MySQL module.

Troubleshooting

- **Private Key Errors:** If the LogAttribute reports `"No Snowflake private key found"`, ensure the Coalesce Environment is configured with valid Snowflake credentials. If utilizing Key Pair auth, the key must be uploaded to the Coalesce Environment settings.
- **401 Unauthorized:** Verify the API Token hasn't expired and that the `Authorization` header includes the mandatory `Bearer` prefix.
- **400 Bad Request:** Validate the `environmentID` in the `ReplaceText` JSON matches the numerical ID of the target environment.